

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
APPLIED CHEMISTRY FOR ENGINEERS [CHM-1003]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Calculate the E.M.F. of the copper – nickel cell at 25°C (4)
A) when $[Cu^{2+}] = 0.02\text{ M}$ and $[Ni^{2+}] = 0.2\text{ M}$ ($E^\circ_{\text{cell}} = 1.10\text{ V}$ at 25°C). Write the cell representation and the cell reactions. Define electromotive force (EMF) of a cell.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Write a brief note on: (4)
(i) Solid state batteries with a suitable example. **1M**
(ii) Internal shorts and thermal runaway on safety concerns with lithium-ion battery. **1M**
(iii) Influence of power density and energy density on battery performance. **1M**
(iv) Limited fast charging and reduced voltage output in NiMH batteries. **1M**

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Give reason: (2)
(i) Proton exchange membrane fuel cells are suitable for use in passenger vehicles. **1M**
(ii) No-maintenance batteries require no addition of water. **1M**

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) (i) Explain the basic structure of MicroOLED's. **2M** (4)
(ii) Explain the Mechanism of degradation of polyhydroxyalkanoates (PHAs) and mention any two of its applications. **2M**
A)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) (i) Calculate the number average, weight average molecular weight and PDI of polymer molecules (4)
with different degrees of polymerization such as 500, 550, 650 and 800 that are mixed in a molecular ratio 1: 2: 3: 4 in a sample of high polymer of styrene (molecular mass = 104). **3M**
(ii) Write the polycondensation reaction between terephthalic acid and ethylene glycol. **1M**

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (i) Explain the principle of Gel Permeation Chromatography (GPC) on Separation Based on (2)
Molecular Size. **1M**
(ii) The intrinsic viscosity $[\eta]$ of a Polymer solution is found to be 0.50 dL/g at 25°C in toluene.

Calculate its viscosity average molecular weight ($K=1.6 \times 10^{-4}$ dL/g and $a=0.9$). **1M**

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3) (i) Discuss zero calibration and span calibration in catalytic bead sensor. **1M** (4)
(ii) Explain antibodies and nucleic acid as bioreceptors in biosensors. **2M**
A) (iii) Explain the principle of flame photometry. **1M**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) (i) Discuss in detail the working principle of minimally invasive continuous glucose monitoring (CGM) sensor. **3M** (4)
(ii) Explain any two types of electrochemical sensors. **1M**

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Explain the role of chemical sensors in water quality monitoring and soil monitoring with suitable examples? **2M** (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) (i) List the countries which record the highest e-waste production and e-waste recycling rate as per the e-waste statistics. **2M** (4)
A) (ii) List the potential health effects of lead and mercury exposure. **2M**

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) (i) Discuss the major health effects faced by workers engaged in e-waste dismantling in Seelampur, India. **1M** (4)
(ii) Explain how repair, refurbishment and second-hand markets can reduce electronic waste. **3M**

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Give reason(s) for the following statements: (2)
(i) Corrosion reduces battery performance by affecting its cycle life. **1M**
(ii) The lead-acid battery does not require separate compartments for anode and cathode. **1M**

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 5) (i) Mention any two applications of pH meter. **1M** (4)
(ii) Draw a neat-labelled diagram of flame photometer? **1M**
A) (iii) A solution of a colored compound has a concentration of 0.008 mol/L. When a light of wavelength 540 nm passes through a 1 cm cuvette containing the solution, the absorbance is measured as 0.90.

- (a) Calculate the molar absorptivity (ϵ) of the compound at 540 nm. **1M**
(b) What will be the absorbance if the concentration is increased to 0.03 mol/L using the same cuvette and wavelength? **1M**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) Explain in detail the basic structure and the working principle of OLED's. (4)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) Mention types of Extrinsic conducting polymers with examples. **2M** (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CHM-1003:APPLIED CHEMISTRY FOR ENGINEERS

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	16
3	3	CO 3	14
4	4	CO 4	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CHM-1003:APPLIED CHEMISTRY FOR ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	9	28(56%)
3	Application	6	22(44%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)